

DFIR Resilience Framework

Research Plan — Questions, Methodology, and Report Structure

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1. Research problem

Determine how the completeness and reliability of digital-forensic evidence affects (a) the accuracy of automated incident reconstruction and (b) the correctness of the incident-response decision derived from that reconstruction, using a reproducible synthetic-incident framework where ground truth is known exactly.

Research questions

- **RQ1 — Degradation curve (core):** How does reconstruction accuracy change as evidence completeness decreases, for the source-removal condition tested?
- **RQ2 — Source importance (core, 2-way):** Between process-creation and network/DNS evidence, which contributes more to correct reconstruction for this attack-chain family?
- **RQ3 — Decision threshold (core):** At what evidence-completeness level does the derived IR decision stop matching the ground-truth-correct decision, and does this happen before or after reconstruction accuracy itself visibly drops?
- **RQ4 — Failure mode (stretch, conditional):** Does degrading both sources at once produce roughly additive accuracy loss, or a non-linear collapse? Only answerable if the combined-degradation stretch goal is reached.

Hypotheses

- **H1 (core):** Reconstruction accuracy decreases monotonically, but not linearly, as evidence completeness decreases.
- **H2 (core, 2-way):** Process-creation/parent-child evidence is more load-bearing than network/DNS evidence for this attack-chain family, because it anchors the causal graph.
- **H3 (core):** IR-decision correctness degrades faster than raw graph-reconstruction accuracy — the system can be “mostly right” about the graph while recommending the wrong response.
- **H4 (stretch, conditional):** Combined degradation (both sources removed simultaneously) produces a measurably super-additive accuracy loss compared to either source removed alone.

Contribution

(1) A small, open, reproducible synthetic-incident generator + evidence + degradation pipeline for one attack family; (2) an empirical resilience curve and a two-way source-importance comparison for that family; (3) a “resilience threshold” metric that ties evidence completeness directly to IR-decision reliability rather than stopping at graph-similarity scores. This is deliberately framed as a small, honest, single-family finding — not a generalizable claim about enterprise DFIR.

Gap

Public DFIR datasets (DARPA OpTC/TC, CICIDS, etc.) provide realistic, complete evidence for detection research, but they don't systematically vary evidence completeness against a known ground truth to study reconstruction robustness, and they rarely connect reconstruction quality to a concrete response decision. Most degradation/robustness work in security targets ML-model robustness to adversarial input, not DFIR-analyst/tool robustness to missing evidence.

What is actually measured

Per experiment run: (1) graph/timeline reconstruction accuracy against ground truth, (2) a confidence score from the reconstruction engine, (3) the IR decision produced, compared against the ground-truth-correct decision — all as a function of a controlled independent variable (which evidence source was removed, at what severity).

2. Evaluation methodology

Metric	What it tells you	In scope
Edge precision / recall / F1	Did reconstruction find the right relationships, without hallucinating extras?	Core — primary metric
Graph edit distance	Single-number structural-similarity score	Stretch
Timeline error	Tests timestamp accuracy specifically	Out of scope (no jitter condition tested)
Attack-stage recall	Coarser, more forgiving than edge-level	Stretch
Confidence calibration	Does confidence correlate with true accuracy?	Stretch
IR-decision correctness	The practitioner-facing metric; answers RQ3	Core

Edge precision/recall/F1 remains the primary metric — simplest to defend, easiest to compute, easiest for readers to interpret. Everything else is secondary and attempted only with slack time.

3. Resilience measurement

Forensic resilience, defined operationally: the range of evidence-completeness values over which reconstruction accuracy and IR-decision correctness both remain above a pre-declared acceptability bar. Suggested bars: $F1 \geq 0.8$ and IR-decision correctness $\geq 90\%$ — decide these before running experiments, not after.

The resilience threshold is the evidence-completeness percentage at which IR-decision correctness first drops below its bar, plotted against the (typically higher) percentage at which raw reconstruction accuracy drops below its own bar. The gap between these two thresholds is the single most interesting result this scope can produce — it operationalizes H3.

4. Experiment design

Variable	Value
Incident families	1
Degradation conditions	2 (one source removed / other source removed) + baseline
Seeds per condition	5 (needed for mean $\pm$ std)

Variable	Value
Approx. total runs	~15 (3 conditions × 5 seeds), plus stretch combined condition

The full undegraded run for the family/seed pair is the baseline/control condition. Report mean $\pm$ std across the 5 seeds in every plot/table — single-run numbers are not defensible.

5. Results the prototype should produce

- Table: baseline (undegraded) accuracy for Family 1 — sanity check before showing any degradation
- Line plot: reconstruction F1 vs. % evidence removed — the primary resilience curve (RQ1/H1)
- Bar chart: accuracy drop for each of the 2 sources removed — the source-importance comparison (RQ2/H2)
- Line plot: IR-decision correctness vs. evidence completeness, overlaid on the F1 curve — shows the RQ3/H3 threshold gap
- Stretch: scatter/calibration plot (confidence vs. actual accuracy)
- Stretch: combined-degradation table vs. sum of individual effects (RQ4/H4)

6. Report plan (short technical report)

Scope and write this as a short technical report, not a conference-length paper with a full related-work section. Keep the structure below proportionate to a report.

- **Title:** “How Much Evidence Is Enough? A Resilience Threshold for Forensic Reconstruction and Incident-Response Decisions Under Evidence Degradation (Single-Family Study)” — keep the finding, not the sample size, as the headline.
- **Abstract:** state the resilience-threshold finding (accuracy vs. IR-decision-correctness gap) as the headline, framed by the small reproducible framework that made it measurable.
- **Introduction:** motivate with the practitioner problem (incomplete/unreliable logs are the norm in real incidents), then state the gap (§1 above) in one paragraph.
- **Related work:** 3–4 sentences citing DARPA OpTC/TC and provenance-graph reconstruction as context — not a literature review.
- **Methodology:** environment/generator, the evidence rule table, the degradation type, the reconstruction rules — condensed from the companion implementation plan.
- **Experiments:** §4 above.
- **Results:** §5's figures/tables, organized around RQ1–RQ3 (RQ4 only if reached).
- **Discussion:** interpret the accuracy-vs-decision-correctness gap; note the practical implication of the source-importance result for log-retention priority, with the caveat that it's a 2-source comparison.
- **Limitations:** §7 below.
- **Future work:** a 3rd/4th evidence source, a 2nd/3rd incident family, other degradation types, combined degradation, real-log validation.
- **Conclusion:** restate the threshold finding and its practical implication, scoped honestly to one incident family and one degradation type.

7. Honest novelty assessment

What already exists: synthetic enterprise attack datasets with full evidence (DARPA OpTC/TransparentComputing, CICIDS); provenance-graph-based incident reconstruction research; general robustness-to-missing-data research in ML.

What this adds: the combination of (a) a ground-truth-known synthetic generator, (b) systematic, reproducible evidence degradation as the independent variable, and (c) evaluation that goes past graph similarity into an actual downstream IR decision. None of these three ideas is individually new; tying them together — even for one incident family and one degradation type — is a small, honest contribution.

Claims to avoid: do not claim the results generalize beyond this synthetic framework, this single incident family, or this pair of evidence sources. The attacker model is a weighted random walk, not real adversary behavior, and the evidence-generation rules are not validated against real Sysmon output. Frame every result as “within this small synthetic study,” which is standard, defensible practice — not a weakness to hide.

8. Threats to validity

- **Synthetic data, unvalidated:** neither the environment nor the evidence has been checked against a real enterprise network or real Sysmon output — state this directly.
- **Simplified attacker behavior:** the weighted-random-walk generator is not adversarial or adaptive.
- **Arbitrary probability values:** action-selection weights are chosen for variety, not calibrated to any real attacker-behavior statistic.
- **Small evidence-source set:** only 2 of a possible 4–5 sources are implemented — the source-importance finding (RQ2) is a 2-way comparison, not a ranking.
- **Single incident family:** findings may not hold for other attack-chain shapes.
- **Small sample size:** ~15–20 runs total — report confidence intervals honestly and avoid overstating statistical significance.
- **Rule-based reconstruction ceiling:** a rule-based correlator plateaus below what a tuned ML approach might reach — a deliberate explainability-over-performance tradeoff.
- **Simulator bias in evaluation:** the same team builds generator, evidence, and (indirectly) knows the reconstruction approach — the no-ground-truth-leakage test (companion plan §7) is the main mitigation.

9. What to say if asked “why only one family / one degradation type?”

Say so directly: the team prioritized a methodologically sound answer to RQ1–RQ3 on a small scope over a methodologically thin answer to RQ1–RQ4 on a large one. This is a defensible, explicit tradeoff — name it as such in the report’s limitations section, not as an apology.